

GeM-CompliAI Platform Guide

AI-Powered Integrated Bid Compliance Verification & Public Procurement Intelligence Platform

Executive Purpose: GeM-CompliAI is a next-generation statutory compliance and risk intelligence engine designed for the Government e-Marketplace (GeM). It automates tender eligibility verification, performs live 8-gateway cross-reconciliation (GSTN, MCA21, Udyam MSME, PAN/ITR, EPFO/ESIC, DPIIT, Debarment Watch, and Make In India local content), and equips Procurement Officers with explainable AI reasoning and cryptographic Class-3 DSC decision signing under General Financial Rules (GFR) 2017.

1. Role-Based Access Control (RBAC) & Dedicated Landing Pages

The platform strictly separates administrative, indenting, and bidding capabilities into three distinct user personas, each routed directly to their authorized workspace upon login:

User Persona	Landing Page	Authorized Capabilities	Security & Governance Scope
Procurement Officer (Competent Authority)	AI Compliance Engine (DOSSIER)	- Bidder Matrix & Cross-Verification - AI Statutory Reasoning & Citations - Class-3 DSC Decision Signing - Audit Vault Logs (SHA-256) - Buyer Desk, Bids/RA, IMS & Analytics	Authorized under GFR Rule 144(i) for multi-portal technical scrutiny and signing binding qualification orders.
Buyer Desk (Indenting Department)	Marketplace Catalog (MARKETPLACE)	- Direct Purchase Orders (GFR 149) - Tender / RFP Drafting & Publishing - Consignee CRAC Management - GeM GPA Escrow Payment Authorizations - Public Bids & Analytics	Indents department requirements, formulates checklists, and manages goods delivery acceptance.
Seller Desk (Registered Bidder / OEM)	Seller & OEM Hub (SELLER_PORTAL)	- Self Pre-Check Verification Tool - Sealed Bid Document Submission - Live Reverse Auction (RA) Bidding - Compliance Certificate Downloads - Direct PO Order Fulfillment	Strictly restricted from viewing competitor dossiers, officer risk indices, or buyer internal audit logs.

2. The 14-Step Automated AI Verification Pipeline

GeM-CompliAI executes an end-to-end 14-step automated verification workflow for every tender and participating bid package:

Step	Workflow Stage & Operational Mechanics
Step 1	Tender Upload & RFP Notice — Procurement Officer / Buyer uploads official tender document in PDF format.
Step 2	AI Rule Parsing & Clause Extraction — NLP parser extracts eligibility criteria, turnover thresholds, past experience, and mandatory certificates.
Step 3	Dynamic Checklist Formulation — Generates structured eligibility rules (Turnover, MSME status, MII local content, ISO standards).
Step 4	Bidder Sealed Package Submission — Sellers submit company profile, GSTIN, PAN, Udyam number, audited financials, and technical spec sheets.
Step 5	Document AI (OCR & UDIN Extraction) — High-accuracy OCR extracts text, detects Chartered Accountant UDIN numbers, and validates dates.

Step 6	8-Government Portal Gateway Query — Automated synchronous lookups against GSTN, MCA21, MSME Udyam, Income Tax PAN, EPFO, DPIIT, and CVC Debarment.
Step 7	Cross-Portal Data Reconciliation — Compares claimed bidder values (e.g. ₹45 Cr turnover) against authoritative portal records (e.g. ₹38 Cr on MCA21).
Step 8	Discrepancy & Fraud Interception — Flags blacklisted entities, expired GSTINs, shell company indicators, and fraudulent MSME certificates.
Step 9	Multi-Factor AI Risk Scoring (0–100) — Computes a weighted statutory risk score across 6 risk vectors (Legal, Tax, Financial, Technical, Debarment, MII).
Step 10	Explainable AI Statutory Citations — Produces plain-English legal explanations citing specific GFR 2017 rules (e.g. GFR 144(i), 151(iii), 175).
Step 11	Compliance Certificate Generation — Creates tamper-proof, QR-coded digital compliance certificates for qualified vendors.
Step 12	Officer Evaluation Review — Officer inspects side-by-side portal proofs, OCR highlights, and automated recommendations.
Step 13	Binding Decision Signing (Class-3 DSC) — Officer applies digital cryptographic signature (DSC) to seal qualification / rejection.
Step 14	Immutable SHA-256 Audit Vault Storage — Every verification, API response, and officer decision is hashed and stored in an immutable audit ledger.

3. Core Platform Modules & User Interfaces

The GeM-CompliAI frontend comprises 9 primary functional modules tailored to government procurement operations:

1. AI Compliance Engine (Bidder Dossier) (Tab: DOSSIER)

Comprehensive side-by-side inspection desk for Procurement Officers. Displays participating bidders, verified turnover from MCA21 vs declared turnover, GSTN filing compliance, MSME classification, and MII local content percentages with instant color-coded risk chips (Low / Moderate / High / Critical).

2. Statutory Evidence & Explainable AI (Tab: AI_EXPLANATION)

Transparent AI reasoning engine providing legal citations for every qualification or rejection recommendation. Cites specific provisions of General Financial Rules (GFR) 2017 (Rule 144, 151, 175) and Public Procurement (Preference to Make in India) Order 2017.

3. Marketplace Catalog & GFR 149 Direct PO (Tab: MARKETPLACE)

Full e-Marketplace catalog supporting GFR Rule 149 Direct Purchase (orders up to ₹25,000) and L-1 lowest verified comparison (orders up to ₹5,00,000). Features instant filtering by MSME, MII Class-1 (≥50% local content), and technical specification comparison.

4. Bids & Reverse Auction (RA) Cockpit (Tab: BIDS_RA)

Live bidding cockpit displaying active tenders, published custom bids, and live Reverse Auctions. Includes dynamic decrement validation, auto-extension countdown timers (+15 minutes if a bid is placed in the final 10 minutes), and live L-1 leader tickers.

5. Buyer Procurement Desk & CRAC Management (Tab: BUYER_CONSOLE)

Buyer order fulfillment registry. Tracks Consignee Receipt and Acceptance Certificate (CRAC) inspections within the mandatory 10-day statutory window. Issuing CRAC releases GeM Pool Account (GPA) escrow payments directly to suppliers.

6. Seller Desk & Self Pre-Check Hub (Tab: SELLER_PORTAL)

Vendor portal allowing registered suppliers to browse open tenders, test their profile compliance against RFP rules prior to formal submission, track bid evaluation status, and download official qualification certificates.

7. Incident Management System (IMS) & Debarment (Tab: IMS)

Statutory dispute and contract breach registry. Manages Show Cause notices, Liquidated Damages (LD) deductions for delayed deliveries, and maintains the National Debarment / Blacklist Watchlist under GFR Rule 151.

8. National Procurement & Compliance Analytics (Tab: ANALYTICS)

Executive macro dashboard presenting total procurement Gross Merchandise Value (GMV), 25% MSME mandate compliance tracking, Make In India penetration rates, and 100% real-time uptime status across all 8 government API gateways.

9. GFR 2017 Rules & Legal Compendium (Tab: RULES)

Interactive reference compendium of Indian public procurement statutory laws, CVC guidelines, MSME Public Procurement Policy, and Make in India orders with clause-by-clause summaries.

4. Authoritative Government Gateway Integrations

GeM-CompliAI connects to 8 official government databases to eliminate dependency on self-declared seller certificates:

Statutory Gateway	Authority / Ministry	Verified Parameters & Data Points	Fraud Protection
-------------------	----------------------	-----------------------------------	------------------

GSTN Gateway	Goods and Services Tax Network	• Active / Cancelled GSTIN Status • Monthly GSTR-3B & GSTR-1 Filings • 3-Year Declared Turnover	Prevents tax defaulting & inactive shell entities.
MCA21 Registry	Ministry of Corporate Affairs	• CIN & Incorporation Validity • Director Identification (DIN) • Audited Balance Sheet Filings	Detects common directorship & collusive cartels.
Udyam Registry	Ministry of MSME	• Micro / Small / Medium Status • Manufacturing / Service Category • Plant & Machinery Investment Cap	Stops fraudulent MSME price preference claims.
Income Tax / NSDL	Central Board of Direct Taxes	• Permanent Account Number (PAN) • ITR Verification & Filing Regularity • Statutory Profit & Loss Proof	Eliminates fake PAN and fabricated balance sheets.
EPFO & ESIC	Ministry of Labour & Employment	• Active Employee Headcount • Monthly ECR Electronic Challans • Social Security Compliance	Verifies genuine technical & operational workforce.
DPIIT Startup India	Dept for Promotion of Industry	• DIPP Recognized Certificate # • Turnover / Experience Exemption • 10-Year Age Eligibility	Ensures valid startup exemptions under GFR 173(i).
CVC Debarment List	Central Vigilance Commission	• Central Blacklisted Entities Watch • Debarment Period & Order # • Offense & Statutory Grounds	Automated instant disqualification under GFR 151.
MII Local Content	Make in India Standing Desk	• Class-1 (≥50%) vs Class-2 (20-50%) • Bill of Materials Location Audit • Statutory CA/Auditor Certificate	Blocks foreign-assembled goods claiming MII status.

5. Cryptographic Security & Technical Architecture

GeM-CompliAI implements defense-in-depth government cybersecurity standards to safeguard sensitive commercial tender bids:

- **AES-GCM-256 Envelope Encryption:** Tender eligibility rules and sensitive price envelopes are encrypted client-side before storage, accessible only via authorized Officer passphrases.
- **Class-3 Digital Signature Certificate (DSC):** Officer decisions are signed with PKCS#7 / X.509 digital signature digests for non-repudiation in judicial or vigilance audits.
- **Immutable SHA-256 Audit Trail:** Every verification lookup, evaluation score, and timestamp is stored in an append-only cryptographic ledger with tamper-evident checksums.
- **Role-Based JWT Authentication:** Secure session isolation with biometric/Aadhaar OTP 2FA simulation for Government NIC accounts.
- **Zero-Knowledge Verification Architecture:** Competitors cannot view competing bids, ensuring strict compliance with GFR Rule 144 confidentiality.

6. Platform Technology Stack

Layer	Technology / Framework	Key Responsibilities
Frontend UI/UX	React 19, Vite, Lucide Icons, GeM Vanilla CSS	Ultra-responsive, government-compliant UI, role-gated navigation, live RA cockpit, and PDF export.
Backend API	FastAPI (Python 3.13), Uvicorn	High-performance asynchronous RESTful microservices for verification, scoring, and report generation.
Database Layer	SQLite3 & SQLAlchemy ORM	Relational schema managing Tenders, Bids, Verifications, Decisions, CRAC orders, and Audit Logs.
AI / NLP Engine	LLM Prompt Orchestrator & OCR Parser	Rule extraction from RFP PDFs, UDIN validation, and multi-factor statutory risk index calculation.
Cryptography	Web Crypto API (SubtleCrypto) & PyCryptodome	AES-256-GCM symmetric envelope encryption, PBKDF2 key derivation, and SHA-256 audit hashing.

7. Local Deployment & Quickstart Reference

- 1. Start Backend Server:**
cd backend && python -m uvicorn main:app --host 127.0.0.1 --port 8000
- 2. Start Frontend App:**
cd frontend && npm run dev
- 3. Access in Browser:**
 - Web Application: <http://localhost:5174/>
 - Interactive FastAPI Docs: <http://localhost:8000/docs>